than one part in a hundred million billion. It's as if you were playing The Price Is Right and Bob Barker gives you ten billion billion dollars and challenges you to purchase products that will cost—cancel, so to speak—all but 189 of the dollars, not a dollar more or less. Coming up with such an enormous yet precise expenditure, without being privy to the exact prices of the individual items, would severely tax the acumen of even the world's most expert shoppers. In string theory, where the currency is energy as opposed to money, approximate calculations have conclusively shown that analogous energy cancellations certainly can occur, but for reasons that will become increasingly clear in subsequent chapters, verifying the cancellations to such a high level of precision is generally beyond our theoretical ken at present. Even so, as indicated before, we shall see that many other properties of string theory that are less sensitive to these finest of details can be extracted and understood with confidence.

This takes us to the third consequence of the enormous value of the string tension. Strings can execute an infinite number of different vibrational patterns. For instance, in Figure 6.2 we showed the beginnings of a never-ending sequence of possibilities characterized by an ever greater number of peaks and troughs. Doesn't this mean that there would have to be a corresponding never-ending sequence of elementary particles, seemingly in conflict with the experimental situation summarized in Tables 1.1 and 1.2?

The answer is yes: If string theory is right, each of the infinitely many resonant patterns of string vibration should correspond to an elementary particle. An essential point, however, is that the high string tension ensures that all but a few of these vibrational patterns will correspond to extremely heavy particles (the few being the lowest-energy vibrations that have near-perfect cancellations with quantum string jitters). And again, the term "heavy" here means many times heavier than the Planck mass. As our most powerful particle accelerators can reach energies only on the order of a thousand times the proton mass, less than a millionth of a billionth of the Planck energy, we are very far from being able to search in the laboratory for any of these new particles predicted by string theory.

There are more indirect approaches by which we could search for them, though. For instance, the energies involved at the birth of the